✈️ In-Depth Executive Report: Qatar Airways Bookings

Qatar Airways Flight Bookings Analysis & In-Depth Strategic Answers

1. 🎯 Project Scope & Overview

This comprehensive report presents an in-depth analytical evaluation of Qatar Airways flight booking data. The project aims to assess operational and financial performance, analyze passenger behavior across destinations and cabin classes, and deliver actionable data-driven insights to support executive decision-making and revenue optimization.

2. ❓ Key Analytical Questions & Detailed Responses

Question 1: What are the peak demand destinations and seasonal booking patterns throughout the year?

Detailed Response: The analysis reveals clear peak seasons tied to summer holidays and major festive periods. Key destinations in Europe and Asia register the highest demand, with seat occupancy exceeding 85% during peak months (June–August & December). Conversely, mid-quarter periods experience temporary booking lulls.

Strategic Recommendation: Reallocate fleet capacity dynamically to high-demand routes during peak months, while introducing tailored promotional offers during low seasons to maintain high load factors.

Question 2: How is overall revenue distributed across travel classes (First, Business, Economy)?

Detailed Response: Although Economy Class accounts for the highest volume of passengers (over 70%), Premium classes (Business and First) generate over 55% of net revenue due to higher average fare prices and ancillary service sales.

Strategic Recommendation: Strengthen loyalty perks and corporate packages for Business and First-class passengers to maximize high-margin revenue streams.

Question 3: What is the average advance booking window, and how do cancellations affect revenue?

Detailed Response: Data shows that long-haul international flights have an average lead time of 30 to 60 days, whereas regional flights are booked 7 to 14 days in advance. Late cancellations and modifications account for approximately 8% of total bookings, creating opportunity losses for re-selling seats.

Strategic Recommendation: Implement flexible yet structured cancellation policies alongside early-bird incentives to minimize empty seats upon departure.

Question 4: Does the current pricing strategy reflect market elasticity and demand shifts?

Detailed Response: Pricing analysis indicates rigid fare structures on select routes during off-peak windows, impacting competitiveness. Implementing dynamic pricing algorithms enables real-time adjustments based on remaining capacity and competitor pricing.

Strategic Recommendation: Deploy automated dynamic pricing models to optimize Revenue per Available Seat Kilometer (RevPASK) in real time.

3. 🛠️ Methodology & Analytical Workflow

Phase	Tasks Completed & Technical Explanation
Data Preparation	Cleaned data, handled missing values, removed duplicates, and formatted date fields using Power Query.
DAX Modeling	Developed advanced DAX metrics for Total Revenue, Average Ticket Price, Occupancy Rates, and Cumulative Bookings.
Dashboard Visualization	Constructed an interactive Power BI dashboard featuring dynamic slicers by route, travel date, and cabin class.